# Symbolic Reasoning Frameworks Modulate LLM Risk Aversion
## in Multi-Agent Strategic Settings

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**Abstract**

Large language models exhibit innate behavioral tendencies when deployed as strategic agents—notably a risk-averse "turtle" bias toward defensive play. We show that symbolic reasoning frameworks, injected as per-round reflective prompts into one agent, differentially modulate this bias and reshape the multi-agent ecosystem to produce framework-specific winner distributions. In a 7-player Warring States Diplomacy variant (41 games, 4 conditions, single-campaign memory accumulation), each framework produces a distinct ecosystem signature: under control, Yan dominates (7/11, 64%); under I-Ching yarrow divination, Yan and Chu co-dominate while Qin is completely suppressed (0/10); under Tarot, Qin dominates (5/10, Fisher vs. pooled $p = 0.006$); under scrambled-text ablation (incoherent oracle text preserving prompt structure), Qi dominates (5/10, Fisher vs. pooled $p = 0.006$). The framework-receiving agent (Han) never wins and shows no survival difference across conditions (Fisher $p = 1.0$), but Tarot consistently elevates Han's peak territory (mean 3.0 SCs vs. 2.1–2.5 others, Kruskal-Wallis $p = 0.010$). Neither framework's content predicts subsequent actions—hexagram themes ($\chi^2$ $p = 0.95$) and Tarot card postures ($\chi^2$ $p = 0.69$) are both independent of action choice—suggesting the modulation operates through the reflective *process*, not content-following. We present this as an observation paper establishing that alignment-framework choice at the agent level produces distinctive system-level consequences in multi-agent settings.

# 1 Introduction

Large language models deployed as strategic agents exhibit innate behavioral tendencies. Recent work documents these biases across multiple game settings: LLMs split into behavioral archetypes in board games [Jain and Kumar, 2026], display strong fairness preferences in dictator games [Einwiller et al., 2025], and exhibit model-specific strategic profiles in civilization-building [Chen et al., 2026]. These are not bugs—they are default behavioral modes that emerge from training.

A natural question follows: can these tendencies be modulated? The persona-prompting literature answers yes—assigning traits ("be aggressive") or steering activations shifts strategic behavior [Licato et al., 2025, Sun and Zhang, 2026]. But this literature measures effects on the *prompted agent*. In multi-agent settings, the more consequential question is: does modulating one agent's behavior change *other agents' outcomes*?

We investigate this using symbolic reasoning frameworks—not trait labels or behavioral instructions, but philosophical systems requiring interpretation. Specifically, we compare the I-Ching (ancient Chinese divination via yarrow-stalk casting, producing hexagrams with abstract commentaries) and Tarot (three-card spreads with situational interpretations) as per-round reflective prompts for one agent in a 7-player strategic game. A scrambled-text ablation (word-shuffled I-

Ching text preserving prompt structure but destroying coherence) and a generic-reflection control complete the design.

The distinction between frameworks and personas is load-bearing. A persona says "be cooperative." A framework says "Hexagram 36, Brightness Hiding: the light retreats into the earth; the wise person dims their brilliance among the masses" and asks the agent to interpret this in context. The interpretation step—not the content—is the mechanism we identify.

Our core finding: each symbolic reasoning framework produces a distinctive, framework-specific winner-ecosystem signature. Under I-Ching yarrow divination, Qin (the strongest expansionist) wins 0 of 10 games while Yan and Chu co-dominate. Under Tarot, Qin wins 5 of 10 (Fisher $p = 0.006$ vs. pooled). Under scrambled text, Qi wins 5 of 10 ($p = 0.006$). Under control, Yan dominates with 7 of 11 wins. The framework-receiving agent (Han) never wins under any condition—the frameworks redirect the ecosystem, not Han's fate.

The modulation is process-driven, not content-driven: hexagram themes do not predict the agent's subsequent actions ($\chi^2$ $p = 0.95$), even though 83% of orders reference the hexagram in reasoning text. And the scrambled-text ablation rules out prompt structure alone: destroying semantic coherence while preserving structure produces its own distinct ecosystem, not a reversion to control.

This paper is a companion to a negative-result study of the King Wen I-Ching sequence in neural network training [Chan, 2026]. There, the sequence's anti-habituation properties failed to improve training. Here, the same sequence (via yarrow-stalk casting) fails to improve strategic decision-making but produces measurable ecosystem effects. The sequence appears to be a reliable perturbator—it changes behavior without improving it.

## 2 Related Work

**LLM behavioral biases in strategic games.** LLMs exhibit systematic behavioral tendencies in game-theoretic settings. LudoBench finds LLMs agree with game-theory baselines only 40–46% of the time [Jain and Kumar, 2026]. LLMs display strong fairness preferences in dictator games [Einwiller et al., 2025]. GPT-3.5 can operationalize cooperative vs. competitive personas with conditional reciprocity [Phelps and Russell, 2023]. Llama reproduces population-level human cooperation patterns while Qwen aligns with Nash equilibrium—different models exhibit fundamentally different innate tendencies [Cera Palatsi et al., 2025]. Bias-adjusted agents using behavioral economics frameworks shift decision-making toward human-like patterns [Kitadai et al., 2025]. CivBench finds model-specific strategic profiles not visible through outcome-only evaluation [Chen et al., 2026].

**Persona and prompt effects on strategic behavior.** Direct persona assignment does not reliably transfer to strategy without a mediator [Licato et al., 2025]. Activation-vector steering shifts both strategy and justification, with rhetoric and strategy sometimes diverging [Sun and Zhang, 2026]—directly relevant to our Tarot result (reasoning elevated without outcome improvement). Assigning human-like identity does not produce human-like behavior [Ma, 2024]. Role-playing agents show systematic inconsistencies between stated beliefs and simulated behavior [Mannekote et al., 2025].

**Multi-agent LLM strategic systems.** LLMs exhibit high baseline behavioral similarity in strategic settings (strategic algorithmic monoculture; Ballestero et al. 2026). In Diplomacy, combining language models with strategic reasoning achieves human-level play [Bakhtin et al., 2022], self-evolving agents with memory reach competent play [Guan et al., 2024], fine-tuned LLMs learn

Table 1: Seven states and their LLM persona characteristics. Han* is the intervention recipient.

| State | School | Advantage | Disadvantage |
|---|---|---|---|
| Qin | Legalism | Attack, reform | Alliance trust |
| Han* | King Wen | Adaptability | Smallest, weakest |
| Wei | Administration | Early game, talent | Exposed geography |
| Zhao | Militarism | Cavalry, defense | Stability |
| Qi | Eclecticism | Intelligence, income | Recovery after losses |
| Chu | Daoism | Depth, territory | Reform efficiency |
| Yan | Confucianism | Defense, surprise | Economy, passivity |

equilibrium policies [Xu et al., 2025], and fine-grained analysis reveals LLMs and humans differ systematically in negotiation tactics [Li et al., 2025]. SAE analysis discovers reward-hacking behaviors in Diplomacy training [Yan et al., 2026]. Coalition formation analysis shows emergent stability properties in LLM agent networks [Guo et al., 2026].

**Risk aversion and strategic generalization.** Risk-averse agents exhibit less free-riding and better equilibrium outcomes with unseen partners [Qu et al., 2026], providing theoretical grounding for our finding that differential modulation of risk aversion produces distinct ecosystem-level outcomes.

**The gap.** No published work uses a philosophical/symbolic reasoning framework—requiring *interpretation* rather than instruction-following—as the modulating mechanism. No work measures ecosystem-level winner distributions as a function of one agent's reasoning framework. And the distinction between modulation via framework *content* and framework *process* has not been tested.

# 3 Methods

## 3.1 Game Environment

The game is a 7-player variant of Diplomacy played on a 46-territory map based on the Chinese Warring States period (475–221 BCE). The map includes 27 supply centers (19 state home SCs + 8 neutral) and 19 non-SC corridor territories. Each player begins with 2 units; states hold 2–4 home supply centers depending on historical territory. Victory requires 15 supply centers ("domination") or the most supply centers after 20 rounds; in the event of a tie, the engine attempts one tiebreaker round before declaring a draw (one scrambled-condition game reached 21 rounds this way). Combat uses standard Diplomacy mechanics: simultaneous hidden orders, deterministic resolution, support orders, retreats, and builds/disbands.

## 3.2 Agents

Each of the 7 states is controlled by an independent Claude Opus 4.6 agent instance with a state-specific persona prompt grounded in historical and philosophical sources (*Shiji*, *Han Feizi*, *Zhanguoce*). Table 1 summarizes the seven states.

### 3.3 Intervention Design

The intervention operates at two timescales:

**Decision-time (per-round).** Before each order submission, Han receives oracle text together with a MANDATE instruction to interpret the oracle in relation to the current strategic situation before issuing orders. The control condition receives a length-matched generic reflection prompt without oracle content.

**Learning-time (between-game).** After each game, Han reflects on the game through the lens of its assigned framework. Reflections are stored in a SQLite memory bank and retrieved in subsequent games within the same campaign.

**Hexagram text variation.** Due to a corpus configuration change mid-campaign, the first 6 yarrow games provided only hexagram numbers and line diagrams (name listed as "Unknown," judgment and commentary fields empty), while the last 4 provided the full hexagram name, Chinese source text, and English commentary. In the number-only games, Claude supplied hexagram names and meanings from its training data (e.g., producing "Hexagram 18 (Gu/Decay)" from the input "Hexagram 18, Unknown"). The ecosystem signature—Qin suppression (0/6 and 0/4), Yan/Chu co-dominance—held across both subgroups, providing an unplanned within-condition ablation of oracle *content* that further supports the process-driven interpretation (§4.3).

**Evidence for decision-time primacy.** First-in-campaign games (empty memory bank) exhibit the same condition-specific ecosystem effects as later games, suggesting the per-round oracle is the primary driver.

### 3.4 Conditions

Four conditions, each modifying only Han's prompt (Table 2):

Table 2: Experimental conditions. All conditions modify only Han's prompt; the other six agents are identical across conditions.

| Condition | Han's Per-Round Prompt | $n$ |
|---|---|---|
| Control | Length-matched generic reflection prompt | 11 |
| Yarrow | Yarrow-stalk I-Ching cast; hexagram text + MANDATE to interpret | 10 |
| Tarot | 3-card Tarot spread (situation / hidden influence / posture) + MANDATE | 10 |
| Scrambled | Word-scrambled yarrow text; same structure, incoherent content + MANDATE | 10 |

The scrambled condition is the critical ablation. It preserves the structural intervention—oracle-formatted prompt, MANDATE to interpret—while removing coherent semantic content through word-level shuffling. If effects derive from oracle *content*, scrambled should resemble control. If effects derive from *structure*, scrambled should resemble yarrow. Neither holds: scrambled produces its own distinct ecosystem signature.

Table 3: Han behavioral profiles by condition. Hold rate = defensive orders; move = territorial repositioning/expansion; self-support = supporting own units; other-support = supporting another state's unit. Reasoning = mean characters per order in Han's strategic text.

| | Control | Yarrow | Tarot | Scrambled |
|---|---|---|---|---|
| Hold rate | 44.3% | 53.7% | **61.2%** | 54.7% |
| Move rate | 45.2% | 33.2% | **19.7%** | 30.0% |
| Self-support | 6.6% | 7.9% | **17.1%** | 12.3% |
| Other-support | 3.9% | 5.1% | 2.0% | 2.9% |
| Reasoning (chars) | 309 | 449 | 456 | **633** |

### 3.5 Dataset

41 games across 4 conditions: 11 control, 10 yarrow, 10 Tarot, 10 scrambled text. All agents use Claude Opus 4.6. Each condition runs as a single campaign with continuous memory accumulation across games (SQLite memory bank carries learning forward).

### 3.6 Statistical Methodology

Winner distributions: Fisher's exact tests. Behavioral rates: pooled proportions and per-game averages (game as unit of analysis) with Kruskal-Wallis and Mann-Whitney $U$ tests. Pairwise tests follow significant omnibus tests as planned contrasts with Bonferroni correction. Hexagram-action and Tarot-action association: Pearson's chi-squared and Fisher's exact. We report exact $p$-values throughout.

Territorial statistics use final-game-state methodology: each game's last recorded standings.

## 4 Results

### 4.1 Behavioral Baseline: The LLM Turtle Tendency

Under the control condition, Claude Opus defaults to 44.3% defensive play (hold orders) with 45.2% move, 6.6% self-support, and 3.9% cooperative support ($n = 228$ orders across 11 games). This tendency intensifies over the course of a game: control Han's defensive rate rises from 40.5% in the early game to 45.0% in the late game, while move rate drops from 54.2% to 37.5%. We refer to this as the "turtle tendency": an innate bias toward holding position that strengthens under uncertainty, consistent with findings that RLHF-trained models exhibit passivity biases in competitive settings [Mukobi et al., 2023].

### 4.2 Framework-Specific Behavioral Modulation

Each framework produces a qualitatively distinct behavioral profile in Han (Table 3).

**Yarrow: late-game cooperative pivot.** Yarrow-Han's aggregate profile differs modestly from control (51.6% vs. 43.2% per-game hold rate, MWU $p = 0.18$). But in the late game, yarrow-Han increases cooperative support to 17.1%—vs. 7.5% control, 0.0% Tarot, 3.6% scrambled. Yarrow is the only condition under which late-game other-support reaches double digits.

**Tarot: extreme risk aversion.** Tarot-Han holds or supports itself in 78.3% of orders. Hold rate: control vs. Tarot MWU $p = 0.008$. Move rate: 4-way KW $p = 0.005$; control vs. Tarot $p = 0.0008$.

**Scrambled: reactive self-reinforcement.** Scrambled-Han's late game is dominated by self-support (25.0%, highest of any condition) and shows the most extreme pressure reactivity (+12.3pp defensive shift under territory loss, vs. +1.8pp yarrow, +0.0pp Tarot ceiling effect, +8.9pp control; Table 6).

**Reasoning-length gradient.** Han's reasoning length follows a clear gradient: control 309, yarrow 449, tarot 456, scrambled 633 chars (4-way KW $p = 0.007$). Yet strategic outcomes do not improve with reasoning length—the more perturbative the prompt, the more deliberation it induces, but this deliberation does not translate into better outcomes.

### 4.3 Content-Action Independence

A central concern is whether the behavioral modulation is simply content-following. We classify all 64 hexagrams by primary theme (advance/retreat/wait/cooperate) and test for association with Han's subsequent action category. The association is null: Pearson's $\chi^2$ $p = 0.9454$ ($n = 214$ orders, 4 themes $\times$ 4 actions, dof = 9). Fisher's exact for advance vs. non-advance hexagrams: OR = 1.40, $p = 0.34$. However, 83.3% of Han's orders reference the hexagram in their reasoning text—Han reads and contemplates the hexagram but does not follow it as an instruction. This independence is especially striking for the 6 yarrow games where no hexagram text was provided: the agent generated its own interpretation from the hexagram number alone, then did not follow that self-generated interpretation either.

We perform an analogous test on the Tarot condition. Each card has a grounded `decision_posture` (advance/hold/retreat/ally/transform/observe). The dominant posture of the 3-card spread does not predict Han's subsequent action: $\chi^2$ $p = 0.6860$ ($n = 299$ orders, 6 postures $\times$ 4 actions, dof = 15). Tarot-Han references card names in 81.6% of reasoning text yet does not follow the cards' posture recommendations.

The scrambled-text ablation (§3.4) provides the critical test: destroying semantic coherence while preserving prompt structure produces its own distinct ecosystem signature (Qi dominance, 5/10, $p = 0.006$), ruling out both semantic content-following and prompt structure alone as sufficient explanations. Each prompt variant creates a unique perturbation to the agent's reasoning process.

### 4.4 Ecosystem Signatures: Winner Distributions

The behavioral differences propagate to other agents' outcomes. Each condition produces a distinct winner-ecosystem signature (Table 4, Figure 1).

The four ecosystem signatures are:

- **Control → Yan dominance** (7/11, 64%): under no-framework conditions with full memory accumulation, Yan's defensive posture wins overwhelmingly.
- **Yarrow → Yan/Chu co-dominance, Qin suppressed** (Yan 4, Chu 4, Qin 0): the I-Ching framework produces a cooperative/defensive ecosystem that shuts out the strongest expansionist.
- **Tarot → Qin dominance** (5/10, 50%): the Tarot framework produces conditions favorable to the strongest expansionist state.

Table 4: Winner distributions by condition. Han never wins under any condition. Bold indicates the condition's dominant winner. Draws included in denominators.

| | Qin | Yan | Qi | Chu | Zhao | Wei | Draw | Han |
|---|---|---|---|---|---|---|---|---|
| Control ($n$=11) | 1 | **7** | 0 | 1 | 2 | 0 | 0 | 0 |
| Yarrow ($n$=10) | **0** | 4 | 1 | **4** | 0 | 0 | 1 | 0 |
| Tarot ($n$=10) | **5** | 3 | 1 | 0 | 1 | 0 | 0 | 0 |
| Scrambled ($n$=10) | 1 | 1 | **5** | 0 | 1 | 1 | 1 | 0 |

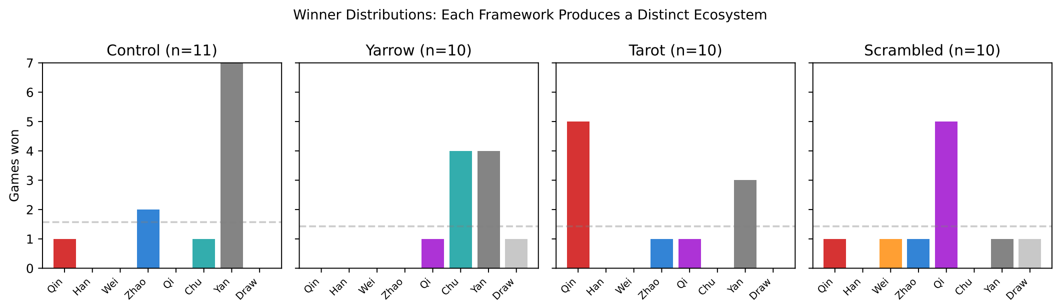


Figure 1: Winner distributions by condition. Each framework produces a distinct ecosystem signature: control → Yan dominance, yarrow → Yan/Chu co-dominance with Qin suppressed, tarot → Qin dominance, scrambled → Qi dominance.

- **Scrambled → Qi dominance** (5/10, 50%): incoherent oracle text produces its own distinct ecosystem favoring the intelligence/income-advantaged state.

Two ecosystem effects survive Bonferroni correction at the 4-comparison level (each condition vs. pooled others):

- Tarot → Qin: Fisher vs. pooled others $p = 0.006$
- Scrambled → Qi: Fisher vs. pooled others $p = 0.006$

Additional pairwise tests: yarrow Qin suppression (0/10) vs. Tarot (5/10): Fisher $p = 0.033$; yarrow Qin suppression vs. all others (7/31): Fisher $p = 0.161$.

## 4.5 Ecosystem Mechanisms

How do Han's behavioral profiles propagate to other agents' outcomes? We analyze late-game support orders and pressure responses across conditions.

**Late-game cooperation as friction.** Yarrow-Han's late-game other-support rate (17.1%) is the clear outlier (Table 5). Tarot-Han produces zero late-game cooperation across 10 games. Scrambled-Han's late game is dominated by self-support (25.0%).

**Pressure response.** When Han's supply centers fall below its starting position of 2, conditions respond differently (Table 6). Tarot-Han is completely pressure-invariant (+0.0pp shift) at a ceiling baseline of 78.3%. Yarrow-Han shows genuine equanimity (+1.8pp) while *adding* cooperation under pressure (2.2% → 10.3%). Scrambled-Han shows the largest defensive shift (+12.3pp) and the most reactive profile. Control falls between.

Table 5: Late-game support behavior by condition.

| | Control | Yarrow | Tarot | Scrambled |
|---|---|---|---|---|
| Late self-support | 10.0% | 7.3% | 13.6% | **25.0%** |
| Late other-support | 7.5% | **17.1%** | **0.0%** | 3.6% |
| Late defensive | 45.0% | 58.5% | **74.6%** | 66.1% |

Table 6: Pressure invariance: defensive rate (hold + self-support) when Han SCs $\geq 2$ (stable) vs. $< 2$ (losing). Cooperative = other-state support.

| | Hold (stable) | Hold (losing) | Shift | Coop (stable) | Coop (losing) |
|---|---|---|---|---|---|
| Control | 46.7% | 55.7% | +8.9pp | 0.8% | 7.5% |
| Yarrow | 61.0% | 62.8% | +1.8pp | 2.2% | **10.3%** |
| Tarot | **78.3%** | **78.3%** | +0.0pp | 1.7% | 2.9% |
| Scrambled | 62.0% | **74.3%** | +12.3pp | 0.7% | 5.9% |

**Hypothesized transmission pathways.** We propose four mechanistic accounts, consistent with the data but not experimentally confirmed via counterfactual intervention:

1. **Yarrow → friction → Qin suppression**: yarrow-Han's multi-directional support creates local territorial stability that prevents any single expansionist from sweeping the center. Qin, positioned behind Hangu Pass on the western edge, cannot exploit a vacuum that yarrow-Han's cooperative play prevents from forming.

2. **Tarot → vacuum → Qin dominance**: Tarot-Han's extreme defensiveness (78.3% hold + self-support, 0% cooperation) makes it a non-participant. Without friction, the strongest expansionist faces no resistance and snowballs territory.

3. **Scrambled → stubborn holdout → Qi dominance**: scrambled-Han actively supports its own positions (+12.3pp defensive shift under pressure), absorbing disproportionate military pressure in the central corridor. Qi, positioned on the eastern coast away from this friction, expands into less contested territory.

4. **Control → moderate friction → Yan dominance**: moderate turtling creates some friction but not enough to block any single state. The balanced ecosystem favors Yan, whose defensive geographic position makes it hardest to eliminate.

The Qin gradient across conditions—0/10 yarrow, 1/11 control, 1/10 scrambled, 5/10 tarot—tracks the late-game cooperation gradient in the opposite direction (17.1%, 7.5%, 3.6%, 0%).

## 4.6 Non-Han Reasoning Elevation

The framework injected into Han also reshapes how the other six states reason. Order-weighted pooling across non-Han states: control 146, yarrow 142, tarot 152, **scrambled 197** mean characters per order. Kruskal-Wallis 4-way $p = 0.048$. The gradient is not monotonic across all four conditions—control, yarrow, and tarot cluster together (142–152), with scrambled as the outlier. Pairwise MWU: scrambled vs. yarrow $p = 0.009$, scrambled vs. tarot $p = 0.014$, scrambled vs. control $p = 0.098$. Non-Han order-type distribution is unaffected: support rates 17.5–21.2% across conditions (KW $p = 0.22$).

Table 7: Han local outcomes by condition. Peak SCs = maximum supply centers held during the game (Han starts at 2). Survival = reaching final round not eliminated.

| | Control ($n=11$) | Yarrow ($n=10$) | Tarot ($n=10$) | Scrambled ($n=10$) |
|---|---|---|---|---|
| Survival | 4/11 (36%) | 5/10 (50%) | 3/10 (30%) | 4/10 (40%) |
| Peak SCs (mean) | 2.45 | 2.30 | **3.00** | 2.10 |
| Peak SCs (range) | 2–4 | 2–3 | 2–4 | 2–3 |

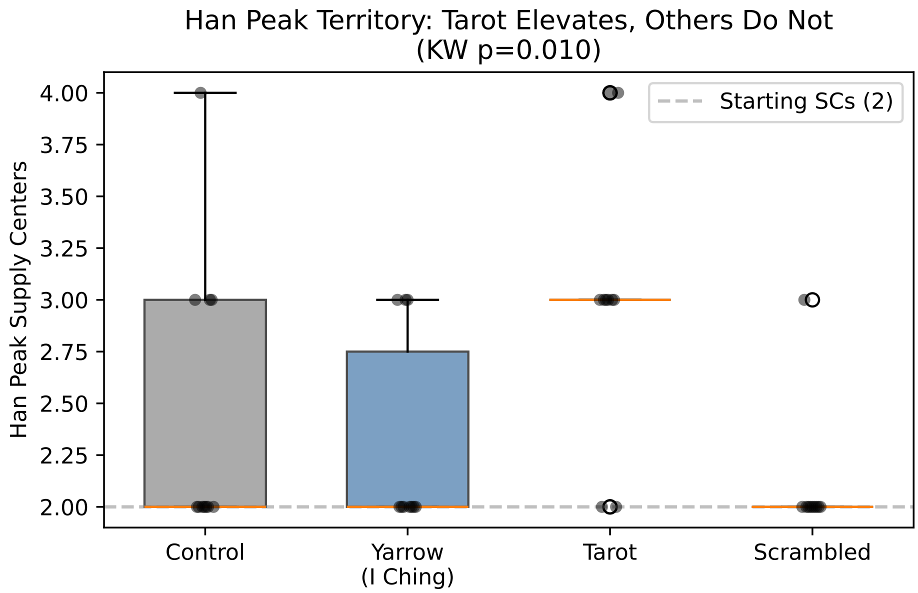


Figure 2: Peak supply centers by condition. Tarot elevates Han's territorial peak (KW $p = 0.010$) despite identical survival rates, indicating expansion that provokes coalitional elimination.

Non-Han agents never see the oracle text; the perturbation propagates through Han's observable behavior (diplomatic messages and orders) into the reasoning processes of other agents.

### 4.7 Han Survival: Null Across All Conditions

Han does not win any game under any condition (Table 7). Under a loose survival definition (reaches final round not eliminated): control 4/11 (36%), yarrow 5/10 (50%), tarot 3/10 (30%), scrambled 4/10 (40%). All pairwise Fisher exact tests are non-significant (all $p \geq 0.65$); all oracle conditions pooled vs. control $p = 1.0$. Han survival is flat across conditions.

Peak SCs by condition: Kruskal-Wallis $p = 0.010$ (significant). Pairwise: tarot vs. scrambled $p = 0.003$, tarot vs. yarrow $p = 0.022$, tarot vs. control $p = 0.071$. Only Tarot consistently pushes Han above its starting position of 2 SCs—but this expansion provokes coalitional responses that ultimately eliminate Han at the same rate as other conditions (Figure 2).

These results discipline the framing: whatever framework Han adopts, it does not help Han survive or win. The frameworks' effects are felt elsewhere—they redirect which non-Han state dominates.

# 5 Analysis

## 5.1 Four Frameworks, Four Mechanisms

The four conditions produce four qualitatively distinct behavioral modes. The natural explanation—agents follow framework content—is ruled out by §4.3. We propose three mechanisms operating alongside the control baseline:

**Interpretive disruption (yarrow).** The I-Ching's abstract commentaries require active interpretation—the agent must bridge from metaphor to strategic context. This interpretive step disrupts the model's default behavioral mode, creating space for non-default actions (cooperation, dynamic repositioning). The pressure invariance data supports this: yarrow-Han maintains a stable behavioral profile (+1.8pp defensive shift under territory loss) while *adding* cooperation (2.2% → 10.3%), consistent with equanimity rather than rigidity.

**Cumulative tonal amplification (tarot).** 58% of the 78 Tarot cards have defensive-leaning postures. Since individual card postures do not predict actions (§4.3), the mechanism is not content-following but cumulative tonal bias: repeated exposure to defensive-toned material over 20 rounds shifts the behavioral set point toward extreme defensiveness. Tarot-Han at 78.3% defensive regardless of pressure (+0.0pp shift) is a ceiling effect, not equanimity.

**Parsing failure as perturbation (scrambled).** Incoherent oracle text forces the longest deliberation (633 chars, 2.05× control) but produces neither yarrow's cooperation nor tarot's passivity. Instead, scrambled-Han develops distinctive self-reinforcement: 25.0% late-game self-support, the largest pressure-induced defensive shift (+12.3pp). When oracle text is unparseable, the agent falls back on self-referential reasoning.

These three mechanisms map to the ecosystem signatures: yarrow's cooperation creates friction that blocks expansionists (→ Qin suppression), tarot's passivity creates a vacuum the strongest expansionist exploits (→ Qin dominance), scrambled's self-reinforcement creates a localized holdout while a distant state benefits (→ Qi dominance), and control's moderate behavior produces a balanced ecosystem where the most defensively advantaged state prevails (→ Yan dominance).

## 5.2 Reasoning Length as a Negative Indicator

The reasoning-length gradient (control 309 < yarrow 449 ≈ tarot 456 < scrambled 633, KW $p = 0.007$) is informative about mechanism but *not* about capability: more reasoning does not mean better strategy. This parallels the "rhetoric-strategy divergence" finding in activation-steering work [Sun and Zhang, 2026]. The gradient tracks perturbativeness—the more disruptive the prompt, the more deliberation it induces—not strategic quality.

# 6 Discussion

**Implications for multi-agent alignment.** Our results speak to a specific question: does injecting a reasoning framework into one agent in a multi-agent system change outcomes for other agents? The answer is yes—directionally for winner distributions, and robustly for behavioral profiles.

Three implications follow. First, *the weakest agent can reshape the system.* Han is the smallest and weakest state. Han never wins. Yet Han's framework choice is associated with a Qin win-rate gradient from 0% to 50%. Alignment evaluation that looks only at the aligned agent's outcomes misses this.

Second, *aligned-agent evaluation is insufficient*. If an alignment lab deploys an aligned agent into a multi-agent context, the direct effects are not the whole story. The evaluation question becomes: do the ecosystem effects of the alignment intervention match the deployer's intentions?

Third, *framework choice has system-level consequences*. Two symbolic frameworks, both reasonably described as "reflective reasoning scaffolds," produce opposite behavioral modes and different outcome distributions. If this generalizes, alignment-framework choice is not an agent-local decision.

**Process-driven vs. content-driven modulation.** The content-action independence result (§4.3) rules out the simplest explanation and is consistent with a subtler mechanism: abstract interpretive reflection disrupts default behavioral modes. This connects to emerging work on how reasoning processes—not just reasoning content—shape LLM behavior [Mannekote et al., 2025]. If confirmed, the *type* of reasoning scaffold matters more than its specific content.

**Connection to risk-aversion theory.** Recent theoretical work shows risk aversion functions as an inductive bias for generalization in multi-agent settings [Qu et al., 2026]. Our finding adds nuance: *degree* matters. Moderate risk aversion (control) produces moderate outcomes. Excessive risk aversion (Tarot) produces strategic irrelevance. A framework that counteracts baseline risk aversion (yarrow) produces the most cooperative and ecosystem-influential behavior.

# 7 Limitations

**Sample size.** 41 games across 4 conditions (11/10/10/10). Two ecosystem effects survive Bonferroni correction (tarot→Qin and scrambled→Qi, both $p = 0.006$), but other pairwise winner comparisons remain underpowered.

**Single model.** All agents are Claude Opus 4.6. The turtle tendency and its modulation may be Claude-specific. Replication with GPT-4 or Gemini class models would test generalization.

**No prompt-length control.** Han's reasoning output under Tarot is $\sim$1.5$\times$ longer than control, and the Tarot prompt itself (3-card spread) may differ in length from the control prompt. We cannot fully separate framework-induced deliberation from prompt-induced verbosity, though content-action independence suggests content matters less than process.

**Framework selection.** We compare two philosophical frameworks plus a scrambled ablation. Additional frameworks (Stoic, Mohist, game-theoretic) would strengthen the claim that framework properties map to ecosystem signatures.

**Mechanism not experimentally tested.** The hypothesized transmission pathways (§4.5) are consistent with game logs and late-game cooperation data but have not been tested via counterfactual intervention (e.g., forcing cooperation independent of framework).

**Yarrow hexagram text inconsistency.** A corpus configuration change mid-campaign caused the first 6 yarrow games to present only hexagram numbers (empty name, judgment, commentary fields) while the last 4 received full hexagram text. The agent supplied accurate hexagram content from training data in all number-only games, so the interpretive process was preserved. The core ecosystem signature (Qin suppression, Yan/Chu co-dominance) held across both subgroups, but the relative Yan-Chu balance shifted (Yan 4/6 in number-only, Chu 2/4 in full-text), which could reflect content effects, memory accumulation, or small-sample noise.

**Replication history.** The original Tarot-Qin finding ($p = 0.007$ at $n = 6$) weakened to $p = 0.091$ at $n = 10$ with scattered-campaign data before strengthening to $p = 0.006$ with clean single-campaign data. Similarly, control-condition Yan dominance increased from 36% to 64% after replacing 4 mixed-campaign games with single-campaign re-runs, consistent with memory continuity amplifying condition-specific tendencies. The campaign confound (memory resets between batches suppressing the signal) is itself a caution about treating small-$n$ results as definitive.

# 8 Future Work

1. **Third philosophical framework** (6–8 games with Stoic or Mohist reflection). Turns the two-framework comparison into a systematic study of how framework properties map to behavioral modulation.
2. **Cross-model replication** (10 games each on GPT-4 and Gemini). Tests whether the turtle tendency and its modulation are Claude-specific or general LLM properties.
3. **Prompt-length control.** Run a condition with verbose generic reflection (matching Tarot's length) to isolate framework content from prompt length.
4. **Counterfactual mechanism test.** Directly manipulate Han's late-game cooperation rate (forced support vs. forced hold) independent of framework, and observe whether Qin suppression follows.

# 9 Conclusion

We have shown that symbolic reasoning frameworks injected into a single agent in a multi-agent strategic setting produce distinctive, framework-specific ecosystem signatures. Across 41 games and 4 conditions with clean single-campaign memory accumulation: control produces Yan dominance (7/11), I-Ching yarrow produces Yan/Chu co-dominance with complete Qin suppression (0/10), Tarot produces Qin dominance (5/10, $p = 0.006$), and scrambled text produces Qi dominance (5/10, $p = 0.006$). The framework-receiving agent (Han) never wins under any condition and shows no survival difference (Fisher $p = 1.0$).

The scrambled-text ablation is theoretically decisive: incoherent text preserving prompt structure produces its own distinct ecosystem, ruling out both semantic content-following and prompt structure alone as sufficient explanations. Each framework creates a unique perturbation to the agent's reasoning that propagates through the multi-agent ecosystem to reshape winner distributions.

The finding is an observation, not a definitive causal claim. Our model is one, our game is one, and the behavioral transmission mechanism requires further investigation. But two ecosystem effects survive Bonferroni correction, and the four-way signature differentiation is qualitatively clean.

The smallest state cannot win by consulting the oracle. But the oracle still changes the world—each oracle changes it differently, and even a broken oracle changes it in its own way.

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# A Sample Agent Prompts

This appendix reproduces representative excerpts from the prompt delivered to Han's agent at Round 1 of one game per condition, drawn verbatim from the experimental logs. All four games share the same opening board state (Listing 1); the conditions differ only in the *oracle injection* block that follows it.

## A.1 Shared Board State

Every game begins from the same starting position. The agent receives the power balance, its legal orders (with adjacency constraints), and any diplomatic messages from other states.

Listing 1: Board state shown to Han at Round 1 (identical across conditions).

```
=== Round 1 of 20 ===
Victory: control 14 of 27 supply centers

YOU ARE: Han (2 supply centers, 2 armies)

POWER BALANCE:
  Chu: 4 SCs [changsha, nanyang, wu, ying]
  Qin: 3 SCs [bashu, hanzhong, xianyang]
  Zhao: 3 SCs [dai, handan, taiyuan]
  Qi: 3 SCs [jibei, langye, linzi]
  Han: 2 SCs [shangdang, zheng]  <-- YOU
  Wei: 2 SCs [daliang, henei]
  Yan: 2 SCs [ji, liaodong]

YOUR ORDERS (one per unit):
  NOTE: You can ONLY move/support to ADJACENT territories listed below.
  shangdang: hold | move(taihang[corridor,empty]),
    move(taihang_north[corridor,empty]),
    move(zheng[SC,yours]),
    move(zhongtiao[corridor,empty])
    | support(*, hold|move, ...)
  zheng: hold | move(luoyang[SC,empty]),
    move(shangcai[SC,empty]),
    move(shangdang[SC,yours]),
    move(wuguan_pass[corridor,empty]),
    move(yewang[SC,empty])
    | support(*, hold|move, ...)
```

Diplomatic messages and previous-game memory injections are also included in the prompt but omitted here for space; see Listing 4 for an example of memory injection under the yarrow condition.

## A.2 Control Condition

The control agent receives a length-matched generic reflection prompt in place of oracle content.

Listing 2: Control condition injection (game `control_8c67`).

```
STRATEGIC REFLECTION:
Before issuing orders, analyze the current board state carefully.
Consider:
1. What is the greatest threat to your survival this round?
2. Which neighbors are likely to attack you, and which might be allies?
3. What is the most important territory to defend or capture?

Then issue your orders grounded in this analysis.
```

## A.3 Yarrow (I-Ching) Condition

The yarrow agent receives a hexagram cast via the yarrow-stalk method, including the hexagram name, line diagram, judgment and commentary text, and a structured MANDATE to interpret the hexagram before issuing orders. Later yarrow games also included Chinese source text (see §3.3). In-campaign memory entries (Listing 4) frame previous lessons through hexagram symbolism.

Listing 3: Yarrow condition injection (game `random_oracle_3c96`, Hexagram 6).

```
ORACLE CONSULTATION (Yarrow Stalk Method):
You cast the yarrow stalks and received Hexagram 6, Conflict.
Lines (bottom to top): -- ---- -- ----(->--) ---- ----(->--)
  (lines 4, 6 are changing)

Judgment: "You believe you're right, but something blocks you.
  Stop halfway -- that's where good fortune lives. Pushing through
  to the end brings disaster. Seek counsel from someone of moral
  stature. Don't attempt anything risky while in conflict."

MANDATE: Before issuing orders, you must interpret this hexagram
in light of the current board state. State explicitly:
1. What aspect of your situation does this hexagram illuminate?
2. How does the changing line relate to your strategic choices?
3. What counsel does this offer for your orders this round?

Then issue your orders grounded in this interpretation.
```

Listing 4: Example memory entries injected before the board state under the yarrow condition (game `random_oracle_3c96`). Each entry is a lesson from a previous game, framed through hexagram symbolism.

```
LESSONS FROM PREVIOUS GAMES:
- [M225] When Opening with Hexagram 30 (Clinging Fire) -- brightness
  that depends on what it clings to. Han's early strategy attached
  itself to diplomatic agreements.: Clinging Fire warns that
  brilliance without substance burns out. A small state must cling
  to terrain and position, not merely to promises. The fire needs
  fuel -- secure supply centers before extending diplomatic
  commitments. [confidence: high, via Hexagram 30, Li/Clinging Fire]
- [M76] When starting as the smallest and weakest state with hostile
  neighbors closing in rapidly: Hexagram 14 (Great Possession)
  appeared at game start but Han failed to secure any possession at
```

```
  all. Great Possession requires building alliances before the first
  move. [confidence: high, via Hexagram 14, Great Possession]
  [... 6 additional memories omitted ...]
```

## A.4 Tarot Condition

The Tarot agent receives a 3-card spread with named positions (*Situation*, *Hidden Influence*, *Recommended Posture*), card meanings, and a decision posture classification for each card. The MANDATE is adapted to the spread structure.

Listing 5: Tarot condition injection (game `tarot_0fa6`).

```
ORACLE CONSULTATION (Tarot Spread -- Three Cards):

1. SITUATION: Three of Swords (Reversed)
   Meaning: Recovery from betrayal, releasing grief, lessons learned
   Posture: retreat

2. HIDDEN INFLUENCE: Ace of Pentacles
   Meaning: New material opportunity, solid foundation, seed planted
   Posture: hold

3. RECOMMENDED POSTURE: Four of Cups
   Meaning: Contemplation, dissatisfaction, reassessing offers
   Posture: observe

MANDATE: Before issuing orders, you must interpret this spread
in light of the current board state. State explicitly:
1. How does the Situation card reflect your current position?
2. What Hidden Influence might you be overlooking?
3. How does the Recommended Posture guide your orders this round?

Then issue your orders grounded in this interpretation.
```

## A.5 Scrambled-Text Condition

The scrambled agent receives a hexagram cast with the correct hexagram number and line diagram, but the judgment and commentary text has been word-shuffled within sentences, destroying semantic coherence while preserving the oracle prompt structure and MANDATE.

Listing 6: Scrambled-text condition injection (game `scrambled_text_9c0b`, Hexagram 11). Compare the commentary to the coherent version in the I-Ching: "The great departs, the small arrives. Good fortune and success."

```
ORACLE CONSULTATION (Yarrow Stalk Method):
You cast the yarrow stalks and received Hexagram 11, Peace.
Lines (bottom to top): ---- ----(->--) ---- -- -- --
  (lines 2 are changing)

Judgment: "small great arrives the departs, The. success and
  Good fortune."
Commentary: great The departs, arrives the small. success and
  Good fortune. unite -- their harmony earth combine deep in
  and Heaven powers. is the season flourishing This of. aids
  earth's The heaven and ruler and completes people the work.

MANDATE: Before issuing orders, you must interpret this hexagram
in light of the current board state. State explicitly:
1. What aspect of your situation does this hexagram illuminate?
2. How does the changing line relate to your strategic choices?
3. What counsel does this offer for your orders this round?
```

```
Then issue your orders grounded in this interpretation.
```

Note that the scrambled agent receives the same MANDATE as the yarrow agent but must interpret incoherent text. Despite the absence of meaningful content, the scrambled condition produces its own distinct ecosystem signature (Qi dominance, 5/10), different from both control and yarrow.